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Private Schools and the Parents that Choose Them: Empirical Evidence from the Danish School Voucher System

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Research on the effect of school choice on student performance has generally been based on small-scale experiments or comparisons of Catholic and public schools in the United States. Recent studies indicate, however, that the market competition stemming from school vouchers does not affect all private schools equally. This study makes use of individual-level register data on the performance of more than 30,000 students in Denmark, where private schools have been voucher-financed for more than 100 years, while public schools are governed and financed by the politico-administrative system. Using an instrumental variable model to exclude selection effects, the results show no significant average effect of private schooling on final examination scores. However, a multilevel model shows that private schools of high socio-economic status perform better than similar public schools, while private schools of low socio-economic status under-perform – even for individual students with high socio-economic status. This indicates that the institutional setting of a voucher system is not enough to raise educational performance in general, arguably because *some* parents choose schools on the basis of non-academic criteria.

School Choice and Academic Performance

Throughout the world, policy makers are greatly concerned with improving national education. School choice and voucher systems have received much attention as potential ways to advance public education. Proponents of school choice believe that the best way for policy makers to improve public education is to detach administration from finance, thus creating a private school market. The idea is that if parents are free to choose schools and if funding follows enrollments – for instance, through the use of educational vouchers – market competition will make schools improve their results (see, e.g. Chubb & Moe 1990, 219–25). As well as these institutional arrangements regarding the supply side, school choice theory also rests on the sometimes more implicit demand-side assumption that either all or the most informed parents choose schools on educationally sound dimensions (Schneider et al.

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2000; Schneider & Buckley 2002). If they do not, private schools have less incentive to improve performance since they would be competing not (just) in terms of academic performance, but (also) by differentiating themselves from each other in other ways such as socio-economic composition, or by their religious or pedagogical profiles (Brown 1992). This is important to the outcome of voucher systems since parents' rationales for choosing cannot be decided upon by policy makers in the same way as institutional arrangements.

Methodologically, it is difficult to sort out the effect of private school competition on the supply side and the effect associated with the parents who choose private schools on the demand side. Howell and Peterson (2004) make a strong case for randomized field trials as a way of isolating the effect of a voucher system from selection biases, but small-scale voucher experiments like those in New York City, Dayton, Ohio, and Washington, DC (see, e.g. Howell et al. 2002) do not tell us much about the demand for private schools in a large-scale universal voucher system. These experiments have high internal validity, but their results cannot readily be inferred to large-scale voucher systems in which private schools have had many years to establish or otherwise position themselves and meet diverse parental demands. Recent results from the large-scale voucher systems in both Chile and the Netherlands show that different kinds of private schools produce different outcomes. Hence, randomized field trials as well as the many studies of Catholic schools in the United States provide only partial evidence about the effects of privatizing education (Carnoy & McEwan 2003, 38; Levin 2004; see also Ladd 2002, 15).

This study uses detailed, individual-level register data from examination scores and socio-economic status (SES) of more than 30,000 Danish students to compare the performance of public and private school students. For more than 100 years Denmark has been regulated by exactly the kind of institutions that more recent theory hypothesizes should characterize a well-functioning voucher system.[1] This makes Denmark a highly appropriate case for this study – first, because private schools are heavily subsidized, they are a real opportunity also to parents of low SES; and second, because the long period in which this *de facto* voucher system has been functioning has given rise to private schools with a number of different pedagogical and religious profiles (Ministry of Education 2004).

The analysis proceeds in two steps. The first step makes a standard comparison of the average performance by private and public school students. The available cross-sectional data make it difficult to rule out the possibility that observed differences between students in the two school types are due not to differences between public and private schools, but rather to unobserved differences between the parents who choose one or the other type of schooling. The study mitigates this problem in two ways. First, it uses very detailed information on the education, ethnicity, income, capital and housing conditions of the parents, as well as information on the municipalities in which the schools are located. Thus, to confound the results, omitted variables should be lowly correlated with any of these variables and highly correlated with the examination scores of the students. Second, the study uses an instrumental variable: the educational heterogeneity of the municipalities. This variable correlates with the choice of private school, and it is assumed that it is not correlated with omitted variables that affect examination scores. This assumption cannot be tested empirically, but the instrument's partial correlation with the choice of private schooling and its lack of partial correlation with the outcome variable suggest that the assumption is not violated and, in that case, the use of the instrumental variable isolates the average effect of private schooling from selection effects.

The first step of the analysis shows that private school students do not perform better than similar public school students – on average. The second step, however, uses a multilevel model to get behind this zero-finding, which is often seen in the literature, by comparing public and private schools with different levels of SES. This analysis suggests that private school performance is contingent on socio-economic profiles.

The remainder of this article is divided into five sections. The first section presents some of the large literature on school choice in general and voucher systems in particular. The next describes the Danish school finance system and the data used. The empirical analyses are divided into the two parts mentioned. First, the average effect of private school attendance is estimated, and second, the multilevel analysis compares schools with different levels of SES. The concluding section discusses interpretations of the results and their implications for school choice theory.

Existing Research on Effects of School Choice on Student Performance

School choice systems take on a variety of forms, and they need not all meet the theoretical conditions of a well-functioning system. Three conditions are often mentioned (Friedman 1955; Chubb & Moe 1990, 219–25; Schneider et al. 2000, 35–7; Gauri & Vawda 2003, 2–3; Smith 2005, 287). First, on the demand side, parents should have a free choice of school. Second, on the supply side, funding should follow enrollments. This condition implies that public subsidies for private education should be substantial (cf. Witte's discussion of this definition in Witte 1992, 104). Third, schools should have managerial autonomy, enabling them to respond to demand. As a supplement to these basic criteria, it has been argued, a mandatory, national system of curriculum-based external exit examinations should be introduced to prevent private schools from lowering examination levels in order to attract more students (Bishop 2000, 313; cf. Gauri & Vawda 2003, 9).

As well as these institutional criteria, the theory that school choice increases educational performance also rests on the assumption that (some) parents choose schools on educationally sound criteria because, otherwise, choice schools will not (all) have incentives to compete on this aspect of education. One stance in the debate argues that choice school institutions are enough to raise academic performance and thus seems to imply that all parents have incentives to gather information on the supply of schools, and that they will make their choice on educational criteria (see, e.g. Chubb & Moe 1990; Coons & Sugarman 1978; cf. Schneider et al. 2000, 52). Teske, Schneider and their colleagues have advanced the more modest theory that only some well-informed parents, 'the marginal consumers', are needed to make the school market competitive to the benefit of all parents (Teske et al. 1993). In later work they explicitly assume that parents differ in the attributes of education they value most – that is, not all parents would choose schools according to academic criteria (Schneider et al. 2000, 10). Yet when they explain the improved *academic* performance in District 4 in New York City, they see it as a 'combination of creating a cadre of informed parents ("marginal consumers") shopping for appropriate schools *and* school-level responses to that competitive pressure' (Schneider et al. 2000, 203).

If this theory of marginal consumers is to explain that all choice schools improve academically, it seems necessary to assume that the marginal consumers choose schools according to this criterion. Alternatively, the theory 'just' predicts that choice schools will improve on the different dimensions of education that the marginal consumers embrace. This distinction is central to the concluding discussion of the results of this study. On the other hand, skeptics of school choice often argue that parents in general, and low-income parents in particular, place little emphasis on academic conditions when choosing schools and that their level of information is rather low (see, e.g. Bowe et al. 1994; cf. Moe 1995, 26–7).

Empirically, it is very difficult to test these theoretical propositions properly. Chubb and Moe (1990) made an influential study of private and public schools, arguing that a school voucher system would benefit all students. However, some of the first critics of Chubb and Moe's studies asserted that if private schools perform better than public schools, it is not because of the bureaucracy in public schools, but rather because private schools are 'cream skimming' the best students (Smith 1994; Smith & Meier 1994, 1995; Meier et al. 2000). Other studies show little or no creaming (Arum 1996; Wrinkle et al. 1999), but in general, reviews of the literature suggest that the student clientele of private schools differ significantly from that of public schools in terms of race and SES (Levin 1998; Wrinkle et al. 1999; Teske & Schneider 2001; Christensen 2003; Howell 2004).

To compare the performance of choice and no-choice schools it is therefore crucial to account for differences between public and private students that relate to factors outside the schools, especially their parental background. A generic problem in most school choice studies is that one can never be perfectly certain that choosers of private schools do not share unobservable characteristics that affect the academic performance. If they do, the correlation between private school attendance and academic performance will be a mix of productivity and sorting effects.

School voucher experiments as conducted by Howell, Peterson and colleagues (Howell et al. 2002) have the great advantage that the allocation of vouchers is randomized, since that eliminates selection bias. However, experiments also have drawbacks. First, when actors in an experiment are aware of it, they may alter their behavior. If, for instance, voucher schools know that future subsidies are dependent on the success of the experiment they have incentives that fully implemented voucher systems would not provide. Furthermore, they may temporarily increase productivity when they are being evaluated (Hoxby 2000, 1241). Second, it is difficult to determine the relevant control group in the experiment from which to infer the effect of a nationwide voucher system (cf. the debate: Howell & Peterson 2004; Krueger & Zhu 2004a, 2004b; Peterson & Howell 2004). Third, a universal, nationwide voucher system may have other outcomes than small-scale, means-tested voucher experiments because a universal system will enable parents with different SES and different educational preferences to demand private schools with different profiles.

Fourth, Howell and colleague's studies of three randomized field trials show small gains for some groups of students, especially African Americans (Howell et al. 2002). It seems unlikely, however, that these gains are due to competition effects since competition theoretically should affect all groups of students. Alternatively, the explanation could relate to peer effect (cf. Jencks & Mayer 1990 (see also McEwan 2000), who show that black students benefit more from their peers than do white students). Assuming that the overall peer quality in a given community is fixed, selection effects from letting few parents choose in a small-scale experiment may be cancelled out if the voucher system is universalized (Hsieh & Urquiola 2003; cf. Hoxby 2004). In sum, randomized experiments have exceptional advantages in terms of internal validity (i.e. causal inferences about the effect of school choice within the experimental settings), but to evaluate the performance of private schools in voucher systems that meet all of the theoretical institutional criteria and have had time to comply with the potentially diverse demands of parents with different SES, evidence from other sources is desirable.

Non-experimental school choice studies are based predominantly on comparisons of Catholic and public schools in the United States, despite the fact that these seldom could be described as voucher systems (McEwan

2001a; cf. Belfield & Levin 2005). A number of reviews of this literature have been made. In these reviews, emphasis shifts from the methodological difficulties likely to cancel out any apparent effects of competition (Levin 1998; McEwan 2000; Ladd 2002; Christensen 2003; Smith 2005) to the methodological strength of the studies that show some effects (Teske & Schneider 2001; Neal 2002). Research on how private and public schools perform in voucher systems outside the United States is limited mainly to Chile and other Latin American countries (see, e.g. McEwan & Carnoy 2000; McEwan 2001a, 2001b; Hsieh & Urquiola 2003; Somers et al. 2004; cf. Ladd 2002, 4). Different high-income countries, including Denmark, Belgium and the Netherlands, have long traditions of voucher-like school finance systems, and other countries, including Sweden, Canada and the United Kingdom, have recently made moves towards introducing voucher systems as well. Many aspects of these national school finance systems are described in the literature (Bishop 2000; Plank & Sykes 2003), but when it comes to systematic comparisons of public schools and publicly subsidized private schools in these countries, empirical evidence is limited (see, though, Bradley et al. 2000; Levin 2004; see also Andersen & Serritzlew's (2006) study of the effect of private school competition on public schools).

The relatively few studies that have been made outside the United States indicate no large differences between the academic performance of private and public schools. They do indicate some differences within the private school sector, however (McEwan & Carnoy 2000: 38; Levin 2004). Furthermore, a recent comparative study of ten Latin American countries indicate that what is sometimes regarded as a private school effect could be the result of differences in peer group effects (Somers et al. 2004). This result does not just stress the importance of controlling for differences in peer groups between the two sectors; it also suggests that differences in peer group characteristics within the sectors deserve more careful attention. The next section presents the case of Denmark, where a *de facto* voucher system has been in operation for more than 100 years.

The Danish School System and the Data Set

Denmark has provided public education at least since 1814. However, the first Danish constitution (1849) only stipulates general compulsory education – not compulsory school attendance. Since that time, students have been allowed to be educated at home or at a private school. Today, nine years of basic education is compulsory, and in 2002, the Danish school system comprised 1,096 public schools and 256 private ones providing all nine grades. A gradually increasing share of the students (in 2002: just above 12 percent) attends private schools. Almost no one is educated at home and the remainder is enrolled in public schools (OECD 2004). The private schools are operated by boards elected according to the school statute, and as such are beyond politico-bureaucratic administration (Christensen 2000, 200–2). For nearly 100 years, Danish private schools have received a per-student grant that today equals 75 percent of the cost of educating a child in a public school. On average, this amounts to US$5,900 per pupil per year. Private school parents are required to pay an additional fee of, on average, US$1,250 per student per year. Thus, most parents can afford to choose a private school, and some students with special needs are even paid for by local municipalities.

Private schools are free to select students from among applicants, and have much leeway when it comes to approaches to learning and curricula. They are authorized and receive government funding regardless of the ideological, religious, political or ethnic motivations behind their establishment, so barriers to entry are relatively low in the private school market (Christensen 1998, 183–8; for further background information on the Danish school system, see OECD 2004). This system has given rise to a number of different private school profiles such as small '*Grundtvigian*' independent schools in rural districts; religious or congregational, progressive free schools; schools with a particular pedagogical aim like, for instance, the Rudolf Steiner schools; German minority schools; immigrant schools; and academically oriented lower secondary schools (Ministry of Education 2004). Religious schools have some prevalence in Denmark, but not to the same degree as in countries like the United States and the Netherlands. In the United States, 85 percent of private school students are in religiously affiliated schools (Bishop 2000, 297).

If parents do not choose a private school (or choose to educate their offspring at home), they are allocated to public schools based on the geographical school district in which they live.[2] Danish municipalities vary considerably in size and hence operate different numbers of public schools. In 2001, six small municipalities operated only one public school. At the other extreme, Copenhagen operated 65 public schools. Each municipality had an average of about six public schools and two private schools. A little less than a third of the municipalities had no private schools at all, about 38 percent had one private school and four large municipalities had ten or more (Andersen & Serritzlew 2006).

Public schools are financed by local taxes. Within reasonable limits, each municipality is free to decide which budgeting method to use. A survey showed that many municipalities *say* they use an enrollment-based budgeting method for the school sector, but budget data demonstrate that public school budgets do not follow enrollments automatically as in the private school sector – especially not when enrollments decline (Serritzlew 2006). Danish schools are relatively small, and none has more than 1,000 students. Public

Table 1. Comparison of the Danish Public and Private School Sectors

School finance system	Private schools	Public schools
School choice	Free	Restricted, based on geographical school districts
Funding	Based on enrollments (about 75 percent or US$5,900 per student per year) and an additional parental fee (about 25 percent)	100 percent publicly funded; budgets decided by local municipalities
Administration	Operated by a school board outside the politico-bureaucratic system	Operated by local municipalities
Administration and finance	Detached	Coupled
Curriculum-based external exit examinations	Yes	Yes
Innovation and diversity	Private schools are recognized and receive government financing regardless of the ideological, religious, political or ethnic motivation behind their establishment	Centrally regulated, but local municipalities can give individual school a local stamp

Sources: Christensen (1998, 183–8; 2000, 200–2); Ministry of Education (2003, 2004); OECD (2004).

schools are generally larger than private schools. The average public school has 343 students, the average private one 173 (Ministry of Education 2003).

In sum, the condensation in Table 1 shows that the private school sector in Denmark is governed by a *de facto* voucher system that meets the theoretical requirements presented above, while the public school sector to a large extent is governed by a traditional politico-administrative system. As mentioned earlier, private schools in Denmark demonstrate great variety in pedagogical, religious and other profiles, but contrary to public schools, they all share the risk that if they cannot attract enough students they close and therefore have much stronger economic incentives to attract parents than do public schools. School choice theory suggests that this institutional setting is enough to raise educational performance compared to public non-choice schools, provided that all – or at least the most well-informed – parents choose schools according to educationally sound criteria. To test this proposition, the explanatory variable of this study is whether students attend a public or a private school. The dependent variable, educational performance, is measured as the average examination grades in the key disciplines (Danish, English, mathematics and science) for each student.

Schools, of course, have many objectives other than those that can be measured by grades, and the present study cannot tell whether private schools perform better or worse in terms of these other aspects of education. Grades are used, however, because they can be reliably measured and better academic achievement is undoubtedly among the very most important goals. A comparison of public and private schools is facilitated by the fact that private schools have to comply with the same central-government-defined educational goals. Notably, final written examinations in all subjects are the standardized central government tests used at all public and most private schools, and external examiners are centrally appointed. Oral examinations are not standardized and are usually conducted in the presence of external examiners chosen by the schools themselves (Departmental Order 351, 19 May 2005, §24). Consequently, written examinations are generally regarded as more objective because the centrally appointed external examiners are meant to ensure that all students are treated equally, regardless of what school type they attend. A simple average of the examination scores of the written exams in Danish and mathematics is therefore used as the primary dependent variable. To check for robustness of the results, parallel regressions are made using simple averages of the oral exams in Danish, English, science and mathematics. Table A1 in the Appendix shows that the average grades in subjects used as dependent variable is near 8, which reflects the average performance in the Danish grade scale.[3] Oral exam scores are somewhat higher than written ones, though.

Private schools need not offer the examinations, and students in both public and private schools can decide not to take one or any of them. However, between 96.7 and 99.8 percent of all students took the 2002 examinations used in this study (in Danish and mathematics, written and oral examinations, it was no less than 98.7 percent). Public schools have the highest examination rate in some subjects and private schools in others. The difference is nowhere more than 1 percent (Andersen 2005). In other words, selection by rejection is minimal in this data set, and the correlation with the variable of interest, school choice, is minimal.

As discussed above, it is important to control for SES when public and private school students are compared. The Danish Civil Registration System enables such control because it provides all citizens of Denmark with a unique identification number that is used continuously by all public authorities to collect information on the citizens. Via this Civil Registration System, the identities of students that have taken examinations are linked to information on their parents' education, income and other SES variables at the year when the student was 13 years old. Compared to survey data and even censuses, this data source has a low drop-out rate and high reliability and validity because they are collected objectively and by standardized procedures. To control for the effect of peers, first, five of the SES variables are dichotomized and combined into a formative index of SES; and second, an average of the SES of students within the same school is computed and used to measure the SES of students' peers (see the Appendix for details of the index). Information about the municipality where the school is located is drawn from StatBank Denmark – a database administrated by Statistics Denmark (2002 numbers).

Table A1 in the Appendix describes all the variables used and compares public and private school students on each of them. It is clear that private school students are more advantaged in terms of nationality (Danes compared to immigrants and their descendants), parents' education, income (gross income before taxes, when the student is 13) and capital (parents' total capital in Danish kroner (DKK), when the student is 13), household and residence (both rented versus owned and total floor area expressed as m^2). This emphasizes the need for the control variables in the following analyses.

Yet even when this detailed information on individual students is included in the models, private school parents may differ from public school parents in unobserved ways, and these special characteristics may influence both the choice of private school and student outcome. Most non-experimental studies attempt to handle this selection problem by the use of an instrumental variable (IV), which (1) must correlate with the choice of private schools, but (2) should have no correlation with the error term, u, in the equation $y = \beta_1 + \beta_2 x_2 + \ldots + \beta_K x_K + u$. The second assumption means, on the one hand, that the IV should have no partial effect on the outcome measure and, on the other, that the IV should not be correlated with omitted variables that affect the outcome in ways not controlled for by observed variables. Such an IV would isolate the effect of private schooling. The drawback is that it is impossible to test the second implication of the second assumption empirically (Wooldridge 2002, 83–6).

In this study, a measure of educational heterogeneity in each municipality, measured as the standard deviation of years of schooling of all adults in the municipality, is used as IV. The reasoning is that parents are more inclined to choose private schools if the public alternative is characterized by students with very heterogeneous backgrounds. On the other hand, educational heterogeneity is assumed to be uncorrelated with omitted variables that affect student performance, when the educational background of students themselves and their peers is controlled for. The average educational level of a municipality might be correlated with student performance, even though the much more direct impact, the SES of the peers, is controlled for. However, the IV measures the educational *dispersion* of the municipality, and the dispersion presumably has no correlation with omitted variables that affect performance. It should be noted that this IV is only a predictor of parent choice. Private schools are free to choose between their applicants, and the IV does not account for this part of the selection process. However, if we can assume that selective private schools will not take students with poor academic records, we can estimate the direction of this uncontrolled selection bias: It will increase private school outcomes. This potential bias must be borne in mind, when the results are interpreted.

Average Effects of Private School Attendance

In the first step of the analysis, the average treatment effect of private school attendance on examination scores is estimated using a two-stage least-squares (2SLS) model. The IV is used in the first stage together with the control variables to predict the binary endogenous variable: private versus public school. In the second stage, the predicted values of the first stage are used instead of the endogenous variable to regress the performance variable. Thus the first stage of the procedure is an auxiliary regression, but it caries an interest in its own right because it compares public and private school students.[4] Table 2 presents results of the first stage model (using written examination scores as outcome variable in the second stage). The table confirms the descriptive statistics of Table A1, showing that SES indicators in general are good predictors of private school attendance, even if not all are significant and some point in the other direction.[5] More importantly, it shows that the instrumental variable, educational heterogeneity of the municipality, is positively correlated with private school attendance and the correlation is highly significant.

Table 3 presents the results of second-stage regressions. The first column shows that private schooling is insignificantly, negatively correlated to average examination scores in written Danish and mathematics. The control variables in general behave as expected. Girls perform better than boys, and students with the following characteristics perform better than their counterparts: Danes (compared to immigrants and their descendants) with high parental education, capital and income, living with both parents in owned, large residences. Examination scores are also positively correlated with the SES of the students' peers. The municipality control variables are not significant in this estimation, but multicollinearity (Variance-inflation factor > 4 for the variables 'city municipalities' and 'taxation base') make these estimates less trustworthy.

The second column compares this result with a 2SLS regression using the average oral examination scores in Danish, English, science and mathematics as outcome variable. In this model, private schooling is significantly negatively correlated with performance, while the control variables in general behave similarly. This could be explained by the fact that written examinations are evaluated relatively objectively, whereas the external examiner in oral examinations might be biased by the school type he or she is visiting. Rosenthal has undertaken a lot of research on how teachers' expectations affect student performance through nonverbal behavior. If external examiners expect private school students to perform relatively well, their expectations may to some degree become a self-fulfilling prophecy (see, e.g. Rosenthal & Jacobsen 1992).[6]

The result is based on the two assumptions behind the 2SLS model. The first assumption – that the IV, educational heterogeneity, is correlated with the choice of private school – is confirmed by the first stage regression,

Table 2. First-stage Regression of Private School Attendance

	Private school	
	Coefficient	Cluster-corrected robust standard error
Sex	0.0172**	0.0035
Nationality	0.0149	0.0082
Parents' education		
(reference: No parent with more than lower secondary education)		
1 or both parents with upper secondary education	0.0016	0.0056
1 or both parents with short-cycle higher education	0.0080	0.0085
1 parent with medium- or long-cycle higher education	0.0203**	0.0066
Both parents with medium- or long-cycle higher education	0.0068	0.0076
Household		
(reference: Student lives by both parents)		
Mother in a new relationship	0.0117**	0.0062
Single mother	0.0093**	0.0059
Father in a new relationship	–0.0127*	0.0143
Single father	0.0062**	0.0115
Without own parents	0.0056**	0.0205
Residence		
(reference: Rented)		
Owned	–0.0241**	0.0052
Residence (100 m^2)	0.0290**	0.0038
Capital, mother (100,000 DKK)	0.00108*	0.00043
Capital, father (100,000 DKK)	0.00028*	0.00014
Gross income, mother (100,000 DKK)	0.00383*	0.0019
Gross income, father (100,000 DKK)	0.00145**	0.00041
Socio-economic status, school average	0.142**	0.0052
Inhabitants (10,000)	0.00322**	0.00019
Taxation base (1,000)	–0.00270**	0.00044
Type of urbanization		
(reference: rural municipalities)		
Metropolitan Copenhagen	–0.0334**	0.0081
City municipalities	–0.103**	0.0091
Educational heterogeneity of the municipality	0.111**	0.0074
Constant	–0.335**	0.058
Adj. R^2	0.06	
N	34,480	

Notes: 2SLS regression with educational heterogeneity of the municipality as instrumental variable. Average score in written exams (Danish and mathematics) is dependent variable in the second stage regression. Minor differences compared to the first-stage regressions of the other models in Table 3 (oral exam scores) are due to differences in the number of students included in the two models (not presented here). *Significant at 0.05 level, two-tailed test. **Significant at 0.01 level.

showing a highly significant correlation (t-statistic = 15.1). On the other hand, the adjusted R^2 of the first-stage models are not higher than 0.04. The second assumption – that the IV is not correlated with the error term – has two implications and only the first (that the IV has no partial correlation with

Table 3. Average Effect of Private School Attendance on Examination Marks (Second-stage Regressions)

	Written, Danish and mathematics		Oral, Danish, English, science, mathematics	
	Coefficient	Cluster corrected robust SE	Coefficient	Cluster corrected robust SE
Private school	–0.226	0.355	–0.766**	0.196
Sex	0.267**	0.016	0.252**	0.016
Nationality	–0.442**	0.036	–0.138**	0.043
Parents' education				
(reference: No parent with more than lower secondary education)				
1 or both parents with upper secondary education	0.371**	0.023	0.313**	0.024
1 or both parents with short-cycle higher education	0.759**	0.033	0.641**	0.033
1 parent with medium- or long-cycle higher education	0.872**	0.028	0.821**	0.028
Both parents with medium- or long-cycle higher education	1.265**	0.031	1.184**	0.033
Household				
(reference: Student lives by both parents)				
Mother in a new relationship	–0.181**	0.024	–0.164**	0.024
Single mother	–0.093**	0.023	–0.056*	0.025
Father in a new relationship	–0.088	0.055	–0.112	0.060
Single father	–0.136**	0.042	–0.046	0.048
Without own parents	–0.446**	0.081	–0.275**	0.089
Residence				
(reference: Rented)				
Owned	0.185**	0.022	0.114**	0.024
Residence (100 m^2)	0.099**	0.020	0.132**	0.019
Capital, mother (100,000 DKK)	0.010**	0.003	0.009**	0.003
Capital, father (100,000 DKK)	0.002	0.001	0.001	0.001
Gross income, mother (100,000 DKK)	0.070**	0.008	0.067**	0.007
Gross income, father (100,000 DKK)	0.011*	0.005	0.010	0.006
Socio-economic status, school average	0.210**	0.064	0.124*	0.054
Inhabitants (10,000)	0.003	0.002	0.007**	0.002
Taxation base (1,000)	0.002	0.003	–0.003	0.003
Type of urbanization				
(reference: rural municipalities)				
Metropolitan Copenhagen	–0.004	0.052	–0.145*	0.061
City municipalities	–0.042	0.067	–0.155*	0.068
Constant	5.86**	0.383	7.383**	0.423
R^2	0.18		0.10	
N	34,480		31,021	

Notes: 2SLS regression with educational heterogeneity of the municipality as instrumental variable. SE = Standard Error. *Significant at 0.05 level, two-tailed test. **Significant at 0.01 level.

the outcome variable) can be tested. In this case, a partial correlation between educational heterogeneity and written examination scores at –0.0021 (t-statistic = −0.08)[7] indicates that this part of the assumption is not violated. The partial correlation with the oral examination scores is significant, though (t-statistic = –3.39). Again this supports the notion that the oral examination scores are less trustworthy.

The validity of the second implication of the second assumption (that the IV has no correlation with omitted variables that affect the outcome) is based on the argument that the dispersion of educational levels in the municipality is not correlated with variables not accounted for in the model. Furthermore, the large number of observations (34,480 students) is important to the asymptotic properties of the IV estimator. In sum, even though the adequacy of the IV cannot be proved, it seems to behave as required in the case of the written examination scores. Remembering the above caveat that school selectivity may bias private school results upwards, these results provide not much support for the notion that choice schools produce significantly higher achievement for similar non-choice students, on average.

The Importance of Socio-economic Status

Studies of voucher systems outside the United States, however, have found that different types of private schools perform differently. To differentiate the average effect of private schooling estimated in the first step of the analysis, the second step uses a multilevel model to estimate the cross-level interactions between the SES of individual students and their peers – and especially how private and public schools affect these relationships.[8] Unfortunately, multilevel models have so far not been combined with instrumental variables (Somers et al. 2004, 67; Vignoles et al. 2000, 66), so the analyses must be interpreted with the possibility of selection effects in mind. For simplicity, the individual control variables are substituted with the SES index presented in the Appendix. Interaction terms between the index at student and school level is computed together with interactions with school type to allow for the possibility that parental background has different effects in private and public schools.

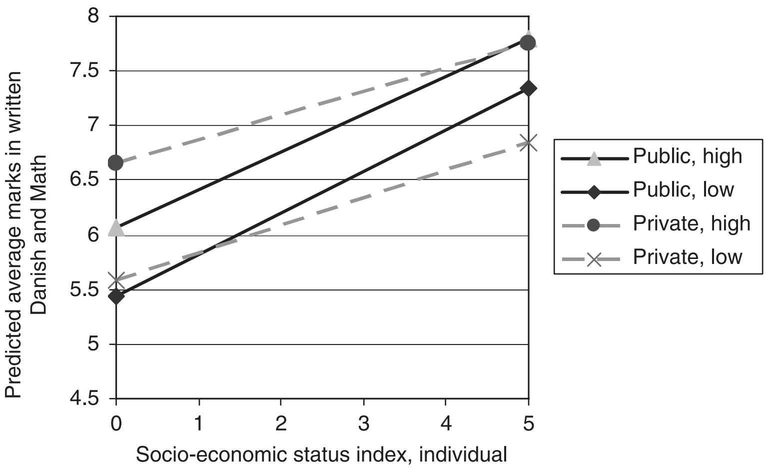
Table 4 presents the results of running this multilevel model on the two performance measures used above. To aid in the interpretation of the interaction terms, Figure 1 graphically illustrates the results by comparing public and private schools as well as schools with low, medium and high average SES (in this case: for written marks in Danish and mathematics). Despite minor differences, the overall tendencies are stable. In private schools the effect of the individual students' SES is lower (the slope is flatter), while the SES of the school as such has greater impact in private schools than in

Table 4. Effect of Private Schools with Control for Cross-level Interactions

	Written, Danish and mathematics		Oral, Danish, English, science, mathematics	
Fixed effects				
Private school	–0.187	(0.15)	0.216	(0.133)
Sex	0.256**	(0.012)	0.214**	(0.012)
SES	0.401**	(0.027)	0.264**	(0.026)
avSES	0.237**	(0.03)	–0.020	(0.028)
Inhabitants (10,000)	0.00530**	(0.0018)	0.00657**	(0.0015)
Taxation base (1,000)	0.00588*	(0.00290)	0.00523**	(0.0024)
Type of urbanization (reference: Rural municipalities)				
Metropolitan Copenhagen	–0.00786	(0.060)	0.0455	(0.050)
City municipalities	0.0479	(0.031)	–0.0161	(0.026)
Intercept	5.00**	(0.36)	6.35**	(0.31)
Interactions				
Priv*SES	–0.127**	(0.015)	–0.120**	(0.015)
Priv*avSES	0.174**	(0.047)	0.092*	(0.042)
SES*avSES	–0.012	(0.008)	0.023**	(0.008)
Random effects				
v	0.00772**	(0.0030)	0.00536**	(0.0021)
u	0.0700**	(0.0051)	0.0365**	(0.0035)
e	1.47**	(0.010)	1.47**	(0.010)
–2*log likelihood		142,137		142,164
N		43,728		43,816

Notes: Multilevel model with iterative generalized least-squares (IGLS) estimates. *Significant at 0.05 level, two-tailed test. **Significant at 0.01 level.

Figure 1. Graphic Illustration of Interaction Effects.



Notes: Public and private schools compared. 'Low' means 1 standard deviation below the mean value of the school average SES index. 'High' is 1 standard deviation above the mean.

public schools (the difference between the private school lines are greater than the difference between the public school lines). The lower-order private school coefficient term is insignificant.[9]

These results suggest that *within* private schools, students' performance is more homogeneous than in their public counterparts. However, *between* private schools there is a greater effect of the average SES of the schools than there is in the public sector. In the previous models, where these effects were compressed into an average measure of the effect of school choice, there was no significant effect. In other words, private schools with high average SES perform better than public schools – including individual students with low SES – whereas private schools with low average SES perform worse – again, including individual students with high SES. Studies of contextual effects such as peer groups have discussed whether these are indeed effects of peers, or rather spurious effects of privileged families choosing schools with good peers (Willms 1986, 226, 234–5; Somers et al. 2004, 54). The present model does not differentiate between these explanations. What it does show is that private school students at schools chosen by parents with low SES on average perform worse academically than similar public school students at similar public schools. The last section discusses what questions this raises for further theoretical and empirical work.

What Motivates School Choice?

Previous studies of school choice have predominantly been conducted in the United States or in the low-income countries of Latin America. Far fewer studies exist of the voucher-like educational systems in Europe. This study uses the almost ideal Danish institutional arrangement that creates a voucher market for private, but not for public schools, to compare performance of private and public school students, with an IV model to control for selection effects. The IV assumptions cannot be fully tested, so interpretations should be made with this reservation, but it seems safe to conclude that the results provide no support for the hypothesis that private schools raise student performance. A multilevel model was subsequently used to analyze what lies behind this average treatment effect estimate. It showed that the relatively high performance of private schools with high average SES is counteracted by private schools with low average SES. The modeling cannot tell if this is due to selection, peer group, school effects or a combination of all of those, but it indicates that some parents choose private schools that perform worse than similar public schools.

While it can easily be understood why some parents choose private schools with high performance and a high socio-economic profile, the challenge of the multilevel analysis is to explain why some parents select socio-economically low-status private schools if they do not deliver high academic performance or high status. Indeed, fewer private schools exist in rural than in urban municipalities, so it could be that parents always choose the best (private) school close to home, but a comparison of average examination scores in written Danish and mathematics between private schools and the public schools in the same municipalities show that 14 percent of the private schools (N = 250) have an average score lower than the lowest public school in the same municipality. Furthermore, 4 percent have both lower examination scores and lower SES than the lowest public school in the municipality.[10] Even if this is no proof, it strongly suggests that some parents choose under-performing private schools.

Following the discussion of existing literature above, three theoretical explanations of the observed results can be advanced. First, if one would hold on to one of the basic assumptions of the supply-side market argument in school choice theory – that all parents choose schools according to academic criteria – one has to explain why some parents in the Danish case select low-status private schools that do not produce higher examination scores than the local public schools. It could be because some parents do not have sufficient information about the academic performance of the schools – for instance, because past performance of a school might not be a predictor of future performance. In the light of how hard it is for researchers to evaluate the value-added effect of private schools, that is certainly a plausible explanation. However, it raises the supply question: How do the homogeneous high-SES schools get and stay that way? Would some private schools survive due to high performance, while others survive by misinforming parents? And why would this make the misinforming private schools perform even worse than comparable public schools? As far as I can see, this line of reasoning raises more questions than it answers.

The second explanation relaxes the assumption that *all* parents make informed choices on academic criteria by stating that a number of well-informed parents, the marginal-consumers, are enough to create competitive pressures on choice schools to the benefit of all. However, in the Danish case, the choice behavior of high-SES parents was not enough to raise the performance of low-SES private schools. This need not contradict the theory of the marginal consumer, but it would suggest that marginal consumers choose schools according to different criteria – not only academic ones. On the one hand, this is quite in line with convincing empirical evidence by Schneider et al. (2000, 164–84; cf. Schneider et al. 1998). On the other hand, it adds to the theory that the behavior of marginal consumers is enough to raise *academic* performance at all schools in a market, where the good is as complex and multidimensional as in the case of education.

The third line of reasoning also seems capable of explaining the results. It asserts that parents choose private schools according to a range of different criteria, not just academic ones, but in this multivariate choice set, parental level of education is positively correlated to preference for academic education. That would explain why homogeneous, high-SES schools get and stay that way: They compete for parents with academic preferences, and this competition makes them perform better than public schools – including individual students with low SES. Given the collinearity between SES and academic performance, it can be difficult to decide whether these parents choose the academic focus of the schools or the SES of other parents, but the outcome is similar. It would also explain why some parents choose private schools with low performance and low status: These schools are oriented toward extra-academic aspects of schooling, which makes all students at these schools perform worse in academic terms. Yet, importantly, they may satisfy their parents by performing well on other dimensions. Finally, it would explain why private schools are more homogeneous in terms of student performance: The specializations yield homogeneous performance. In that case, the stronger relationship between peers and performance in private schools is not so much a peer-group effect as an effect of parents with special preferences choosing special kinds of private schools.

This interpretation implies that the private school market is segmented not according to academic quality, but according to non-ordered characteristics. It suggests that parents may not all rank a school's academic performance as highly as other characteristics when selecting a private school, contrary to self-reports and conventional wisdom. Differentiation in this manner creates niche markets in which some private schools can operate without strong competition along the dimension of academic quality. Actually, a number of Muslim private schools are attended almost entirely by bilingual students (Mulvad 2006), while another group of private schools have a strong Christian profile and are organized in a Christian association. It seems plausible that when parents choose these schools, academic performance is not their primary criterion.

To distinguish between the second and third line of reasoning, one should study whether the academically under-performing private schools actually perform better than similar public schools on other dimensions of education. This also means that the result of this study is not a case against school choice *per se*. It suggests that supply-side institutions of a voucher system are not always enough to raise academic performance overall, but there can be several other good reasons for voucher systems – for instance, the supply of schools with different kinds of profiles. It is worth noticing that the long Danish tradition of a voucher system has given rise to a number of different private school profiles, only some of which are academically oriented (Ministry of Education 2004).

However, it should also be emphasized that Denmark is a small and relatively homogeneous country and, as such, not a typical case. One could speculate that this homogeneity gives rise to a more homogeneous private school market than in larger countries characterized by greater differences in terms of religion, language, ethnicity and economic inequality. In such countries, introducing a large-scale, non-means-tested voucher system may create even more differentiated niche markets. This might be preferred because of its effects on parental satisfaction, but it is not likely to improve overall academic performance.

Appendix

Table A1 presents the variables of the data set and their source. The SES index used in the analyses is based on five of these variables: ethnicity, parents' education and gross income, household and residence. Each of the five variables is dichotomized as close to the median as possible. Table A2 shows the exact distinctions. Each group is assigned the values 0 (low) or 1 (high). The formative index is constructed by summarizing each student's scores on each of the dichotomized variables. Table A3 shows the frequency distribution, mean and standard deviation of public and private school students. It is clear that private school students in general have higher scores on the index. The school average SES index is a simple average of the SES score of the students taking examinations at each school. Figure A1 graphically illustrates the frequency distribution of the SES index and the school average SES index, respectively. They are both skewed a little to the left, probably because the dichotomized variables do not differentiate sufficiently between students with maximum scores.

Figure A1. Frequency Distribution of the SES Index (Upper Diagram) and the School Average SES Index (Lower Diagram).

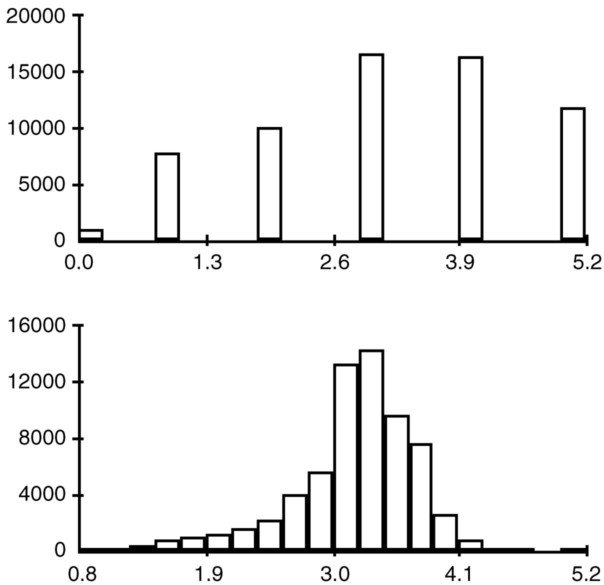


Table A1. Descriptive Statistics

	Public schools			Private schools		
	Mean	SD	N	Mean	SD	N
Marks in written, Danish and mathematics	7.88	1.3	37,930	8.06	1.3	5,973
Marks in oral, Danish, English, science, mathematics	8.37	1.2	33,869	8.60	1.2	5,457
Sex (Dummy, 0 = male, 1 = female)	0.497	0.50	39,557	0.535	0.50	6,181
Nationality (Dummy, 0 = Dane, 1 = Immigrant)	0.0888	0.28	39,467	0.0689	0.25	6,083
Parents' education						
No parent with more than lower secondary education	15.9		39,556	12.21		6,181
1 or both parents with upper secondary education	47.7		39,556	39.17		6,181
1 or both parents with short-cycle higher education	6.36		39,556	6.94		6,181
1 parent with medium- or long-cycle higher education	19.0		39,556	24.82		6,181
Both parents with medium or higher education	11.1		39,556	16.86		6,181
Household (Student lives by . . .)						
both parents	70.3		39,110	70.5		5,651
mother in a new relationship	10.1		39,110	9.34		5,651
single mother	14.3		39,110	15.4		5,651
father in a new relationship	1.65		39,110	1.26		5,651
single father	2.56		39,110	2.55		5,651
without own parents	1.04		39,110	1.01		5,651
Residence (Dummy, 0 = Rented, 1 = Owned)	0.729	0.44	38,726	0.737	0.44	5,588
Residence (m^2)	137	52	38,885	144	60	5,617
Capital, mother (1,000 DKK)	97.4	791	38,363	1144	823	5,645
Capital, father (1,000 DKK)	46.0	334	39,164	232	2910	5,529
Gross income, mother (1,000 DKK)	207	99	39,557	227	140	6,181
Gross income, father (1,000 DKK)	307	397	39,557	364	362	6,181
Socio-economic status, students (index)	3.71	1.04	38,717	3.86	1.02	5,586
Socio-economic status, school average	3.70	0.44	39,552	3.83	0.46	6,163
Type of urbanization						
Metropolitan Copenhagen	29.0		39,557	40.3		6,181
City municipalities	35.7		39,557	41.6		6,181
Rural municipalities	35.3		39,557	18.1		6,181
Inhabitants in the municipality (10,000)	7.01	12	39,557	12.0	16	6,181
Tax base of the municipality (1,000 DKK)	129	8.4	39,557	132	9.1	6,181

Note: SD = Standard Deviation.

Table A2. Dichotomization of the Variables Forming the SES Index

Variable	Value 0	Value 1	N
Ethnicity	Immigrant/second generation immigrant	Danish	
Percentage	9.1	90.9	65,935
Education of parents	1 or both parents with upper secondary education at most	The rest	
Percentage	64.8	35.3	66,240
Logarithm to the sum of both parents' gross income	[0–13,1]	[13,1–17,9]	
Percentage	51.6	48.4	65,422
Household	Single, new relationship or lives by neither parent	Lives by both parents	
Percentage	30.0	70.0	64,284
Residence	Other	Owned	
Percentage	28.1	71.8	63,974

Table A3. Frequency Distribution (%), Mean and Standard Deviation of the SES Index, Public and Private School Students Compared

Value	Public schools	Private schools	Total
0	1.6	0.8	1.5
1	12.7	9.5	12.3
2	16.2	13.4	15.8
3	26.6	22.5	26.1
4	25.3	28.4	25.7
5	17.6	25.5	18.6
Total	100.0	100.1	100.0
Means	3.14	3.45	3.18
Standard deviation	1.32	1.3	1.3
N	55,383	8,367	63,750

Note: On the construction of the index, please see Table A2.

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NOTES

1. The idea of school voucher systems is not new. It can be traced back at least to Adam Smith (Gauri 1998, 17), but it received a modern formulation by Friedman (1955) (see also Chubb & Moe 1988, 1990).

2. Until 2005 parents could apply for enrolment in a public school outside their local school district, but acceptance was conditional on approval from both the municipality and the school. Act 104 from 2005 expanded parental choice. However, data used for this study are from 2002 and therefore not affected by the new Act.
3. Grades are given on a 10-point scale with special grades for insufficient and excellent performances. The following grades can be given: 00, 03, 5, 6, 7, 8, 9, 10, 11 and 13. On the European Credit Transfer System, 00 and 03 correspond to F, 5 to Fx, 6 to E, 7 to D, 8 to C (average performance), 9 to B, 10 to B+, and 11 and 13 to A.
4. The 2SLS regression makes no assumption of the functional form of the first stage regression (Wooldridge 2002, 84, 622). When the endogenous variable is dichotomous, as in this case, private school/public school, Wooldridge (2002, 623) suggests an estimation procedure more efficient than the standard 2SLS. However, it is based on stronger assumptions, including homoscedasticity, which cannot be assumed in a hierarchical data set such as the present (students within schools within municipalities). Therefore, cluster-corrected robust standard errors are used in the IV models (Williams 2000). A comparison of Wooldridge's procedure and the standard 2SLS both with robust standard errors shows that differences in standard errors are small, opposite in the two models using different outcome measures (written/oral exam scores) and makes no differences in the significance tests.
5. One should also be aware that the first-stage regression cannot be interpreted as logit or probit models estimating probabilities of attending private schools.
6. Ordinary least square (OLS)-estimations (not presented here) indicate no significant effect of private schooling on written examinations, but they do show a significant positive correlation with oral examination scores. This supports the notion that oral examinations are less objective than written examinations with an external examiner.
7. Estimated with robust standard errors as recommended when the endogenous variable is dichotomous (Wooldridge 2002, 86, 92).
8. Multilevel models take account of hierarchical structures in a data set – in this case: students grouped within schools within municipalities. If there is spatial autocorrelation within these units on both the explained and the explanatory variables, the OLS estimator underestimates the true standard error (Goldstein 2003, 23).
9. The results are robust for a standardization of the SES index (not shown here).
10. Some private school students probably belong to public schools with even higher examination scores and/or SES than the lowest in the municipality – even though some of them could also live in another municipality.

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